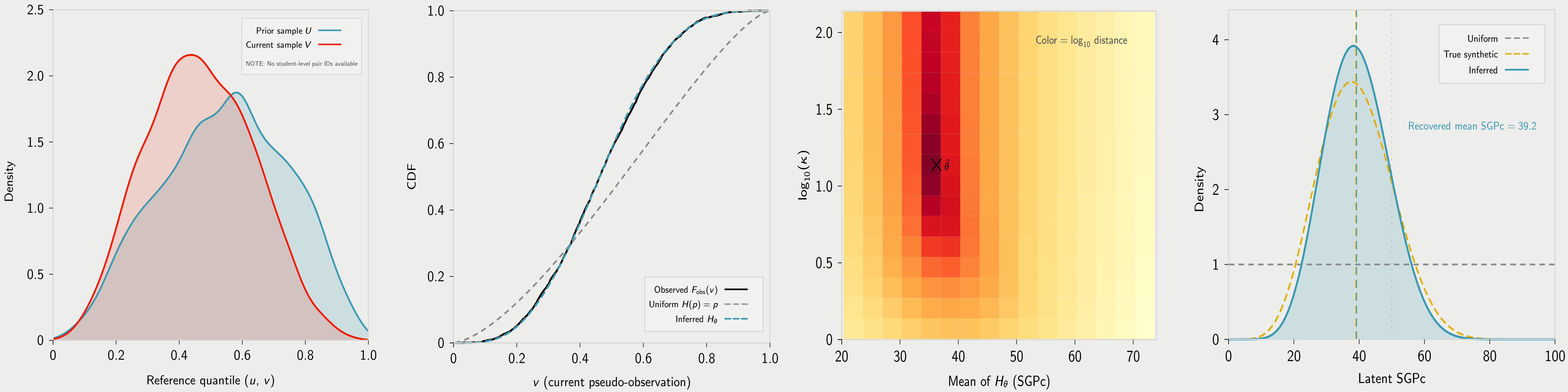
Stochastic Growth: From observed *V* to inferred ***SGPcFlow*** growth regime

Flow A → B → C → D | $W_1(F_{\mathrm{obs}}, F_{\boldsymbol{\theta}})$: uniform = 0.0915, inferred = 0.003 (96.7% reduction)

Synthetic low-growth case (true mean SGPc 39), baseline t-copula kernel (ρ = 0.72, df = 8)



A. Independent cross-sectional inputs

The starting point is two independent cross-sectional samples: prior-grade reference percentiles $U = F_X^{\mathrm{ref}}(x_i)$ and current-grade reference percentiles $V = F_Y^{\mathrm{ref}}(y_j)$.

Crucially, there is *no student-level linkage* between these samples—we cannot match any u_i to its corresponding v_j. This is exactly the situation faced by international assessments such as TIMSS, where Grade 4 and Grade 8 cohorts are sampled independently.

The distributions of U and V are the only observables; all subsequent inference proceeds from these marginals alone.

B. The analytic identity

Given a candidate growth regime H_θ and the baseline transition kernel $F_0(v|u)$ from the fitted copula, the **analytic identity**

$$F_\theta(v) = \mathbb{E}_U\big[H_\theta\big(F_0(v|U)\big)\big] = \frac{1}{n}\sum_{i=1}^{n} H_\theta\big(F_0(v|u_i)\big)$$

predicts the current-grade CDF *exactly*—no Monte Carlo simulation is needed. The uniform regime $H(p) = p$ (grey dashed) systematically misses the observed CDF, while the inferred $H_{\hat{\theta}}$ (teal dashed) closes the gap.

C. Minimising the distance

We search over candidate parameter vectors θ = (mean, κ) of the Beta growth regime, computing the Wasserstein-1 distance

$$W_1\big(F_{\mathrm{obs}},\ F_\theta\big)$$

at each grid point. The heatmap shows $\log_{10}$ distance: dark regions are far from the observed CDF; light regions are close.

A two-stage procedure—coarse grid scan followed by L-BFGS-B refinement—locates $\hat{\theta}$, the parameter vector that minimises the distance between the predicted and observed marginal CDFs of V.

D. The recovered growth regime

The estimated growth regime $H_{\hat{\theta}}$ (teal) is a Beta density on [0, 1] representing the distribution of latent conditional growth percentiles (SGPc). It closely matches the true synthetic regime (amber dashed), recovering both the mean and spread.

The regime separates the subgroup's *growth occupancy*—where students actually fall in the conditional growth distribution—from the baseline *dependence template* encoded by the copula kernel $F_0(v|u)$. A regime centred below 50 indicates growth that is systematically lower than the reference population.